

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		appropriate scales (1) all points plotted correctly (1) (tolerance ± 1 square)		2		2	2	2
		(ii)		straight line extrapolating back to y-axis (1) $\Delta T = 44 - 17 = 27$ (1) where vertical scale covers 0-40 extrapolation back to y-axis is not possible so mark as follows line of best fit drawn (1) $\Delta T = 41 - 17 = 24$ (1)			2	2	2	1
		(iii)		mass of solution = 380g (1) $q = -mc\Delta T = 380 \times 4.18 \times 27 = -42\,886.8$ J (1) $n(\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}) = \frac{320}{136.09} = 2.35$ mol (1) $\Delta H = \frac{-42.9}{2.35} = -18.26$ kJ mol ⁻¹ (1) ECF possible $\Delta H = -16.22$ using $\Delta T = 24$ from part (ii)		4		4	2	1
		(iv)		heat is lost to the surroundings / sodium ethanoate does not fully crystallise			1	1		1
	(b)			award (1) for each reactant $\text{CH}_3\text{COOH} + \text{NaHCO}_3 \rightarrow \text{CH}_3\text{COONa} + \text{CO}_2 + \text{H}_2\text{O}$ award (2) for correct equation using Na_2CO_3 $2\text{CH}_3\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COONa} + \text{CO}_2 + \text{H}_2\text{O}$		2		2		